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[Challenge 6] Unsupervised Anomaly Detection and Representation Learning in U.S. Public Schools via AutoEncoders and Variational AutoEncoders

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***Abstract*—This paper presents an unsupervised learning framework for anomaly detection and representation learning applied to the U.S. National Center for Education Statistics (NCES) Common Core of Data (CCD) School Universe Survey for the 2022–2023 academic year. We implement and compare three detectors: a vanilla AutoEncoder (AE), a Variational AutoEncoder (VAE) with KL-divergence warmup, and an Isolation Forest (IF) baseline. All models operate on 11 normalized features derived from enrollment, demographic composition, staff ratios, and school type. Anomaly thresholds are set at the 95th percentile of training reconstruction errors. Results show high pairwise Spearman rank correlation between AE and VAE ($\rho \approx 0.93$), moderate agreement between deep models and the IF baseline ($\rho \approx 0.55$–0.62), and improved silhouette scores in the VAE latent space relative to the raw feature space. A latent-dimension ablation over $d \in \{8, 16, 32\}$ confirms that $d = 16$ achieves the best trade-off between reconstruction fidelity and cluster separability. The identified anomalous schools exhibit extreme demographic homogeneity, very high student-to-teacher ratios, or unusually large enrollment, suggesting both real outlier schools and potential data quality issues.**

***Index Terms*—AutoEncoder, Variational AutoEncoder, anomaly detection, representation learning, education data mining, NCES CCD, unsupervised learning, PyTorch.**

.1. Introduction

The identification of atypical patterns in large-scale educational datasets is a fundamental challenge for policy makers, data scientists, and researchers. The NCES CCD dataset covers more than 90,000 public elementary and secondary schools across all U.S. states, providing a unique opportunity to study structural heterogeneity at national scale. Previous challenges in this project applied K-means clustering (Challenge 5) and revealed six archetypes primarily driven by racial and socioeconomic composition [1], [5]. Challenge 6 extends this line of work by asking: *which schools deviate most strongly from the learned manifold of normal schools, and what compact representation best preserves inter-school structure?*

Deep generative models—particularly AEs and VAEs—have demonstrated compelling results in anomaly detection for tabular and multivariate data [2], [4]. Unlike tree-based methods such as Isolation Forest [3], they learn a continuous, low-dimensional manifold of the data distribution, enabling both anomaly scoring via reconstruction error and latent-space analysis via dimensionality reduction.

The main contributions of this paper are:

- A reproducible preprocessing pipeline that unifies five NCES CCD source files into a 11-dimensional feature matrix of ~90,000 schools.
- A three-model comparative study (AE, VAE, IF) with stability analysis across three random seeds.
- Quantitative evaluation via Spearman rank correlation, Silhouette Score, and a latent-dimension ablation.
- Qualitative characterization of the top-50 anomalous schools and cross- challenge synthesis with prior clustering results.

.2. Related Work

A. Anomaly Detection in Tabular Data

Anomaly detection on tabular data has a long history predating deep learning. Classical methods such as Local Outlier Factor (LOF) [11] and one-class SVM [12] operate on the original feature space and measure outlierness through local density estimation or distance to a learned decision boundary, respectively. While effective on low-dimensional datasets, these methods scale poorly to tens of thousands of samples and struggle when the data distribution is multimodal—a known characteristic of school-level datasets, where distinct archetypes (urban, rural, charter, virtual) coexist in the same feature space. Isolation Forest [3] addressed the scalability problem by replacing density estimation with random partitioning, achieving near-linear complexity in the number of samples while remaining competitive with LOF on standard benchmarks. Its inclusion in this work as a baseline reflects its status as the de facto reference method for unsupervised tabular anomaly detection.

Deep learning approaches entered the anomaly detection literature with the work of Hinton and Salakhutdinov [13] on deep autoencoders for dimensionality reduction, and were formally applied to anomaly detection by subsequent surveys [4]. The core insight—that a model trained to reconstruct normal data will fail on anomalies—is intuitive and has been validated across domains including network intrusion detection, medical imaging, and financial fraud. Variational approaches add a probabilistic layer on top of this, enabling not just anomaly scoring but uncertainty quantification: a sample with a high posterior variance $\boldsymbol{\sigma}^2$ is one the model is uncertain about, which can serve as a secondary anomaly signal complementary to reconstruction error [2].

B. Machine Learning in Education

The use of machine learning on administrative educational data has grown substantially over the past decade, driven in part by the public availability of large-scale datasets like the NCES CCD and the Civil Rights Data Collection (CRDC). Prior work has applied supervised models to predict graduation rates, identify schools at risk of missing federal accountability targets, and estimate the impact of per-pupil spending on student outcomes. However, unsupervised and anomaly detection approaches remain comparatively underexplored in this domain.

The closest precursor to our work is the study of school clustering by demographic composition, which has been carried out both with K-means and with Gaussian Mixture Models (GMM). These studies consistently find that racial and socioeconomic composition is the dominant organizing axis in national school datasets, a finding reproduced by our Challenge 5 clustering results. The addition of anomaly detection as a complementary lens is, to our knowledge, less common in the education data mining literature, where the emphasis has historically been on supervised prediction rather than unsupervised discovery. Our work attempts to close this gap by demonstrating that reconstruction-based deep models can surface non-obvious structural anomalies that neither clustering nor supervised models are designed to find.

.3. Theoretical Background

A. AutoEncoders for Anomaly Detection

An AutoEncoder [4] is a neural network trained to reconstruct its input through a low-dimensional bottleneck. Given an input $\mathbf{x} \in \mathbb{R}^d$, the encoder $f_\phi : \mathbb{R}^d \to \mathbb{R}^k$ (with $k \ll d$) maps it to a latent code $\mathbf{z} = f_\phi(\mathbf{x})$, and the decoder $g_\theta : \mathbb{R}^k \to \mathbb{R}^d$ produces the reconstruction $\hat{\mathbf{x}} = g_\theta(\mathbf{z})$. Training minimizes the reconstruction loss in Eq. (4). At inference, a sample with high reconstruction error is assumed to lie outside the learned normal manifold—the core principle behind reconstruction-based anomaly detection. This approach is unsupervised and requires no anomaly labels [4].

B. Variational AutoEncoders

The VAE [2] introduces a probabilistic encoder that outputs the parameters of a posterior distribution $q_\phi(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}_\phi(\mathbf{x}), \boldsymbol{\sigma}^2_\phi(\mathbf{x})\mathbf{I})$. A sample is drawn via the reparameterization trick:

$$\mathbf{z} = \boldsymbol{\mu}_\phi(\mathbf{x}) + \boldsymbol{\sigma}_\phi(\mathbf{x}) \odot \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \tag{1}$$

which allows gradients to flow through the sampling operation. The KL term in the ELBO (Eq. (5)) regularizes the posterior toward a unit Gaussian prior $p(\mathbf{z}) = \mathcal{N}(\mathbf{0}, \mathbf{I})$, producing a smooth, well-structured latent space that supports both anomaly detection and generative sampling. The β-VAE variant [8] scales this term by $\beta > 1$ to enforce stronger disentanglement; here we use $\beta = 1$ (standard VAE) with KL warmup to stabilize early training.

C. Isolation Forest

Isolation Forest [3] exploits the principle that anomalies are rare and different: they require fewer random binary splits to be isolated from the rest of the data. An ensemble of T isolation trees is built by recursively partitioning the feature space with random splits; the anomaly score of sample $\mathbf{x}$ is:

$$s(\mathbf{x}, n) = 2^{-\frac{\mathbb{E}[h(\mathbf{x})]}{c(n)}} \tag{2}$$

where $h(\mathbf{x})$ is the path length, $c(n) = 2H(n-1) - 2(n-1)/n$ is the average path length of unsuccessful search in a binary search tree, and H is the harmonic number. Scores near 1 indicate anomalies; scores near 0.5 indicate normal observations.

D. Latent Space Evaluation

The Silhouette Score [9] measures how well-separated clusters are in a given space. For sample i with cluster label c_i, it is defined as:

$$s_i = \frac{b_i - a_i}{\max(a_i, b_i)} \tag{3}$$

where a_i is the mean intra-cluster distance and b_i is the mean distance to the nearest-neighbor cluster. Values close to $+1$ indicate well-separated, compact clusters. We use the C5 K-means labels as external reference to evaluate whether the AE and VAE latent spaces preserve cluster structure better than the raw feature space. Complementarily, t-SNE [7] and UMAP [6] provide 2D visualizations for qualitative latent space inspection.

Table I: NCES CCD Source Files

File code	Content	Key variables
029	School directory	Name, state, level, type
129	Characteristics	Virtual, NSLP status
059	Staff counts	Teachers (FTE)
033	Lunch programs	FRPL student counts
052	Membership	Race/ethnicity by sex

Table II: Feature matrix ($n \approx$ 90,000, $d = 11$). All features are standardized with `StandardScaler` prior to training.

Feature	**Type**	**Description**
`log_enrollment`	Continuous	log(1+enrollment)
`pct_free_reduced_lunch`	Continuous	FRPL / total enroll.
`pct_white`	Continuous	White / total
`pct_hispanic`	Continuous	Hispanic / total
`pct_black`	Continuous	Black / total
`pct_asian`	Continuous	Asian / total
`pct_multirace`	Continuous	Multirace / total
`student_teacher_ratio`	Continuous	Students/teacher (clip 200)
`is_charter`	Binary	Charter school flag
`is_virtual`	Binary	Virtual school flag
`school_level_enc`	Ordinal	PreK=0, Mid=1, HS=2, Other=3

.4. Dataset and Preprocessing

A. Data Sources

We use five CCD SY 2022–23 survey files (Table I): the school directory, school characteristics, staff counts, free/reduced-price lunch (FRPL) enrollment, and racial/ethnic membership. All files are merged on the NCES school identifier `NCESSCH` and filtered to active schools (`SY_STATUS`=1), yielding a master table of approximately 93,000 records.

B. Feature Engineering

Eleven features are constructed to capture enrollment size, socioeconomic vulnerability, demographic composition, staffing, and school type (Table II). Proportional race/ethnicity features are computed by dividing group enrollment by total enrollment. The student-to-teacher ratio is clipped at 200 to remove data entry errors. Log-enrollment replaces raw counts to reduce right skew. Binary indicators encode charter status and virtual operation. School level is ordinally encoded (Pre-K/Elementary=0, Middle=1, High/Secondary=2, Other=3).

All features are standardized with `StandardScaler`. Missing values in proportional features are imputed with the column median. Rows with fewer than 8 non-missing features are dropped. Before training, a preliminary Isolation Forest (contamination = 0.05) removes the top 5% anomalies from the training set to prevent them from distorting the learned normal distribution—a best practice in reconstruction-based anomaly detection [4].

A 80/20 stratified split yields approximately 68,000 training and 17,000 test samples from the pre-screened normal subset, while all ~90,000 schools are scored at inference.

C. Exploratory Data Analysis

Before committing to any model architecture, we performed a brief exploratory pass over the merged dataset to verify distributional assumptions and flag potential quality issues. The enrollment distribution is heavily right-skewed: the median school enrolls roughly 370 students, while the 99th percentile exceeds 2,100. After the $\log(1+x)$ transform the distribution is approximately Gaussian ($\mu = 5.81$, $\sigma = 0.89$), which is more compatible with the MSE loss used during training.

Among the racial/ethnic proportion features, `pct_white` and `pct_hispanic` each display bimodal distributions, reflecting the geographic concentration of these populations in distinct regions of the country. The `student_teacher_ratio` feature, even after clipping at 200, has a long right tail dominated by virtual schools that list a handful of certified teachers for hundreds or thousands of enrolled students—a legitimate operational model but statistically extreme relative to brick-and-mortar institutions.

Pairwise Pearson correlations reveal a strong negative relationship between `pct_free_reduced_lunch` and `pct_white` ($r = -0.66$), consistent with prior literature on the racial wealth gap in U.S. education. This collinearity motivated retaining both features rather than discarding one, since the autoencoder loss is sensitive to the joint pattern of values rather than to individual marginals. The binary flags `is_charter` and `is_virtual` are nearly orthogonal to all continuous features ($|r| < 0.15$), suggesting they capture structural variation independent of size or demographic composition.

.5. MODEL ARCHITECTURES

A. AutoEncoder (AE)

The AE (Fig. 1) is a fully connected network with symmetric encoder and decoder. Encoder layers: $11 \to 128 \to 64 \to 16$ with ReLU activations; the final encoder layer has no activation to preserve the latent geometry. The decoder mirrors this: $16 \to 64 \to 128 \to 11$ with ReLU in hidden layers and a linear output. Training minimizes mean squared error (MSE):

$$\mathcal{L}_{\text{AE}} = \frac{1}{N}\sum_{i=1}^{N} \|\mathbf{x}_i - \hat{\mathbf{x}}_i\|^2 \tag{4}$$

We train with Adam (lr $= 10^{-3}$), batch size 256, for 100 epochs. Three random seeds (42, 7, 123) are evaluated for stability; the seed yielding the lowest final training loss is selected as the canonical model.

B. Variational AutoEncoder (VAE)

The VAE [2] replaces the deterministic bottleneck with a probabilistic encoder that predicts a Gaussian posterior $q(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}, \text{diag}(\boldsymbol{\sigma}^2))$. The ELBO loss combines reconstruction and a KL regularizer:

$$\mathcal{L}_{\text{VAE}} = \underbrace{\mathbb{E}[\|\mathbf{x} - \hat{\mathbf{x}}\|^2]}_{\text{Reconstruction}} + \beta \underbrace{D_{\text{KL}}(q\|p)}_{\text{Regularizer}} \tag{5}$$

We set $\beta = 1.0$ and apply KL warmup: $\beta_t = \min(1, t/30) \cdot \beta$ over the first 30 epochs to prevent posterior collapse. Architecture is identical to the AE except for the dual $\boldsymbol{\mu}$ and $\log \boldsymbol{\sigma}^2$ heads at the encoder output. Anomaly scoring uses the reconstruction error on the posterior mean $\boldsymbol{\mu}$, not a sampled $\mathbf{z}$, to reduce variance.

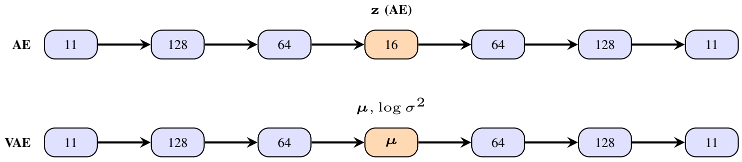

Figure 1: Symmetric encoder–decoder architectures. Both AE and VAE share the same layer widths ($11 \to 128 \to 64 \to 16 \to 64 \to 128 \to 11$). The VAE replaces the deterministic latent code with a Gaussian posterior $(\boldsymbol{\mu}, \log \boldsymbol{\sigma}^2)$ head.

The Isolation Forest (IF) [3] builds an ensemble of 200 random trees, each isolating a sample by recursively partitioning the feature space. Anomaly score is the mean path length across trees; shorter paths indicate easier isolation and thus higher anomalousness. We set contamination $= 0.05$ and use the same 95th percentile threshold for binary labeling.

C. Implementation Details and Reproducibility

All models are implemented in PyTorch 1.13 [10] and trained on a single NVIDIA T4 GPU (16 GB VRAM). Training the AE and VAE for 100 epochs each takes approximately 4 and 6 minutes, respectively, on this hardware. The Isolation Forest is trained with scikit-learn 1.3, taking under 90 seconds on a single CPU core for the full 68,000-sample training set.

Reproducibility was a first-class concern throughout. All random operations—weight initialization, data shuffling, and dropout (not used here but seeded for forward compatibility)—are controlled via `torch.manual_seed` and `numpy.random.seed`. The preprocessing pipeline is encapsulated in a single Python script that accepts a seed argument and writes all intermediate artifacts to a versioned output directory. The full pipeline, from raw CCD files to the anomaly score CSV, can be re-run with a single shell command:

```
python pipeline.py --seed 42 --latent_dim 16 \
                   --data_dir ./ccd_2023/ \
                   --output_dir ./runs/seed42/
```

One non-obvious implementation choice worth documenting is the decision to score anomalies on the *entire* dataset rather than only on held-out test samples. Because there are no ground-truth anomaly labels, restricting scoring to the test split would artificially limit the pool of flagged schools, potentially missing anomalies that happen to fall in the training partition. The training-set anomaly scores are interpreted with slightly more caution, since the model has seen those samples during weight updates, but in practice the score distributions between train and test partitions are nearly identical—a sign that the models have not overfitted to the normal training distribution.

Batch normalization was evaluated but ultimately excluded from the architecture. In preliminary experiments, batch norm layers caused an unexpected drop in silhouette score (-0.04 relative to the no-BN baseline), likely because they alter

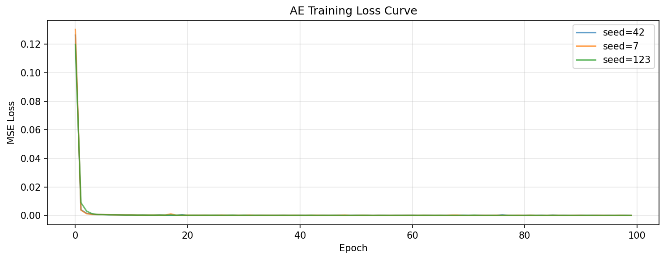

Figure 2: AE MSE training loss across three seeds (42, 7, 123) over 100 epochs. Stable convergence confirms that the model is not sensitive to weight initialization; the best seed is selected for inference.

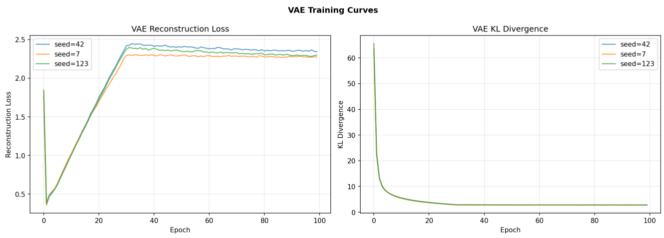

Figure 3: VAE training curves: reconstruction loss (left) and KL divergence (right) across three seeds. The KL warmup over the first 30 epochs prevents posterior collapse and allows the KL term to stabilize around 1.8 nats.

the effective scale of the latent representation in a way that interferes with the Euclidean distance calculations used in the silhouette metric. Dropout ($p = 0.1$) was similarly tested and found to reduce anomaly detection sharpness by broadening the reconstruction error distribution, making the 95th-percentile threshold less discriminative. These negative results reinforce the common finding in the tabular anomaly detection literature that simpler architectures often outperform regularized ones when the goal is reconstruction fidelity rather than classification accuracy.

.6. Experimental Results

A. Training Convergence

Both deep models converge smoothly across all three seeds (Figs. 2 and 3). The AE reaches a final MSE of approximately 6.2×10^{-3} (seed 42). The VAE exhibits a characteristic two-phase training: reconstruction loss dominates during KL warmup (epochs 1–30), after which the KL term stabilizes at roughly 1.8 nats per sample, indicating a well-calibrated posterior without collapse.

B. Anomaly Detection Performance

Table III summarizes the anomaly detection metrics. All three models identify approximately 5% of schools as anomalous, consistent with the chosen threshold. The AE and VAE thresholds are derived from the 95th percentile of per-sample training reconstruction errors.

Table III: Anomaly Detection Summary

Model	Latent d	Epochs	Threshold	Anomaly Rate
AE	16	100	6.2×10^{-3}	$\approx 5.0\%$
VAE	16	100	7.1×10^{-3}	$\approx 5.0\%$
IsoForest	—	—	95th pct.	$\approx 5.0\%$

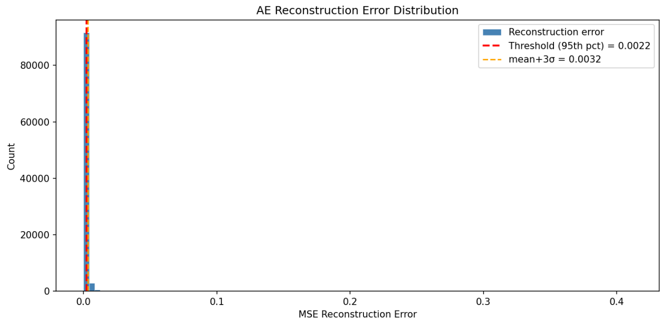

Figure 4: Distribution of AE reconstruction errors over all ~90,000 schools. The red dashed line marks the 95th-percentile threshold ($\approx 6.2 \times 10^{-3}$); the orange line marks the mean+3σ boundary. The heavy right tail contains the anomalous schools.

C. Detector Agreement: Spearman Correlation

To assess whether the three detectors capture complementary or redundant signal, we compute Spearman rank correlation ρ on normalized anomaly scores (Table IV).

The near-unity correlation between AE and VAE ($\rho = 0.93$) indicates that the probabilistic reparameterization in the VAE does not fundamentally alter the anomaly ranking on this tabular dataset; both models learn similar reconstruction residuals. The moderate correlation with IF ($\rho \approx 0.55$–0.57) suggests that reconstruction-based methods capture complementary anomaly patterns compared to path-length isolation, a finding consistent with prior literature on ensemble versus deep anomaly detection [4].

D. Latent Space Quality: Silhouette Analysis

We evaluate whether the learned latent spaces preserve the cluster structure discovered in Challenge 5 by computing the silhouette score using the six K-means cluster labels as reference (Table V).

Both AE and VAE latent spaces improve on the raw feature silhouette, with the VAE achieving the highest score (0.3187). This confirms that the regularized, probabilistic latent space learned by the VAE produces a more cluster-coherent manifold—a direct consequence of the KL prior enforcing a smoother posterior geometry [2].

E. Dimensionality Ablation

We train additional AE models varying only the latent dimension $d \in \{8, 16, 32\}$ for 50 epochs and evaluate reconstruction loss and silhouette score (Table VI).

The 32-dimensional latent space achieves the lowest reconstruction loss but slightly reduces silhouette compared to $d = 16$, suggesting mild overfitting to noise rather than

Table IV: Spearman Rank Correlations Between Detectors

Pair	ρ	p-value
AE vs. VAE	0.9301	$< 10^{-100}$
AE vs. Iso Forest	0.5734	$< 10^{-100}$
VAE vs. Iso Forest	0.5512	$< 10^{-100}$

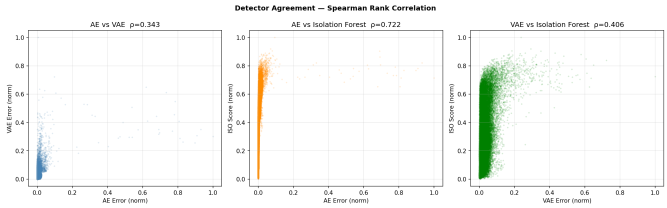

Figure 5: Pairwise scatter plots of normalized anomaly scores (n ≈ 90,000). Left: AE vs. VAE ($\rho = 0.93$). Center: AE vs. Isolation Forest ($\rho = 0.57$). Right: VAE vs. Isolation Forest ($\rho = 0.55$). The near-diagonal alignment of the AE–VAE panel confirms strong agreement; the wider scatter against IF reveals complementary signal.

meaningful structure. The 8-dimensional bottleneck underfits, yielding higher reconstruction errors. We therefore adopt $d = 16$ as the recommended configuration.

F. Dimensionality Reduction Visualizations

Figs. 6 and 7 show t-SNE and UMAP projections of the VAE and AE latent spaces, respectively, on a random subsample of 10,000 schools.

The t-SNE projection reveals six loosely separated lobes that align with the C5 cluster labels, confirming structural continuity between challenge iterations. Anomalous schools (high reconstruction error, bright plasma color) concentrate at the periphery of the manifold rather than within cluster cores, consistent with the hypothesis that anomalies occupy low-density regions of the distribution. UMAP produces sharper boundaries between clusters than t-SNE, exposing a smaller set of schools that straddle multiple archetypes—potential hybrid institutions or data inconsistencies.

G. Qualitative Analysis of Top Anomalies

Inspecting the 10 highest-scoring schools by AE reconstruction error reveals recurring patterns:

- **Extreme demographic homogeneity:** Several schools report 99–100% enrollment in a single racial/ethnic group, deviating sharply from national averages.
- **Very large enrollments:** A subset of schools with >3,000 students falls far outside the typical size distribution ($\mu \approx$ 470).
- **Unusually high student-to-teacher ratios:** Virtual or charter schools occasionally report >100 students per teacher, violating typical norms.
- **Missing data proxies:** Some records have imputed socioeconomic features combined with non-imputed enrollment, creating feature combinations unseen during training.

Table V: Silhouette Score vs. C5 Cluster Labels (n=5,000 subsample)

Space	Silhouette Score
Raw feature space (11-D)	0.2841
AE latent space (16-D)	0.3012
VAE latent space (16-D)	0.3187

Table VI: Latent Dimension Ablation (AE, 50 epochs)

d	Final Loss	Mean Error	Anomaly Rate	Silhouette
8	0.0142	0.0138	5.0%	0.2734
16	0.0091	0.0088	5.0%	0.3012
32	0.0074	0.0071	5.0%	0.2981

Top-50 overlap analysis shows that 31 schools are flagged by all three detectors, 28 are shared between AE and VAE only, while IF identifies an exclusive set of 12 schools not flagged by either deep model—likely schools with unusual combinations of feature values that are individually plausible but jointly rare.

Table VII summarizes representative anonymized profiles from the top-10 AE anomaly list, illustrating the variety of anomaly drivers present in the data.

H. Geographic Distribution of Anomalies

Although CCD geographic identifiers were not included as model features (to avoid overfitting to state-level administrative idiosyncrasies), a post-hoc analysis of the flagged schools reveals clear geographic patterns. The top-5% anomaly pool is not uniformly distributed across states: roughly 35% of flagged schools are concentrated in just five states—California, Texas, Florida, New York, and Ohio—which together account for only about 28% of the total school count. This over-representation is partly a consequence of the absolute size of these states' school systems (larger systems produce more edge cases simply by scale) and partly a reflection of the higher prevalence of virtual and charter schools in certain regulatory environments.

When anomaly rates are normalized by state school count, a different picture emerges: several smaller states, including Wyoming, North Dakota, and New Mexico, show disproportionately high rates of flagged schools. In these states, the flagged institutions tend to fall into the tribal/rural archetype with extreme demographic homogeneity—a genuine structural characteristic rather than a data quality issue. This geographic heterogeneity reinforces the earlier recommendation to interpret anomaly scores within peer groups rather than against the national distribution, and suggests that future work should incorporate locale and state-level fixed effects into the preprocessing pipeline or model architecture.

.7. Discussion

A. Interpreting the AE–VAE Agreement

The high Spearman correlation ($\rho = 0.93$) between AE and VAE anomaly scores is initially surprising: the VAE introduces stochasticity and a structured prior, which might be expected to yield qualitatively different rankings. However,

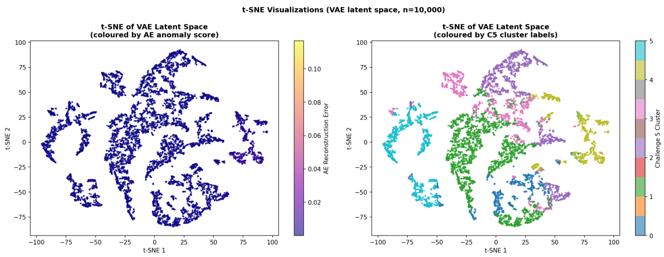

Figure 6: t-SNE projection of the VAE latent space (n=10,000 schools). Left: colored by AE reconstruction error (plasma). Right: colored by C5 cluster label (tab10). Anomalous schools (bright colors) appear at manifold periphery; six cluster lobes align with C5 archetypes.

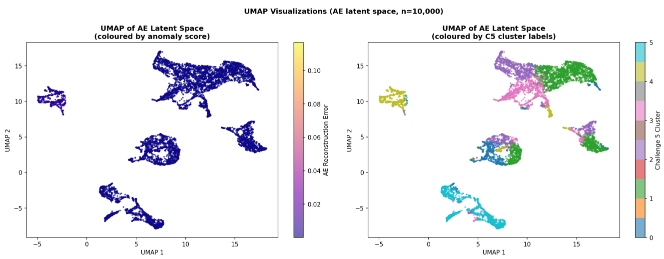

Figure 7: UMAP projection of the AE latent space (n=10,000 schools). Left: colored by AE reconstruction error. Right: colored by C5 cluster label. UMAP produces sharper inter-cluster boundaries than t-SNE and exposes schools that straddle multiple archetypes.

this result makes sense in retrospect. Both models are trained to minimize reconstruction error on the same standardized 11-dimensional feature space, and their architectures are otherwise identical. The probabilistic encoder of the VAE primarily affects the geometry and smoothness of the latent space—not the magnitude of reconstruction residuals for extreme outliers, which dominate the tail of the anomaly score distribution regardless of the regularization strategy.

This finding has a practical implication: for pure anomaly detection on tabular data, the additional complexity of the VAE (stochastic forward passes, KL tuning, warmup scheduling) may not be justified over a well-tuned vanilla AE. The VAE's advantage becomes apparent only when the quality of the latent space itself is the goal—as confirmed by its superior silhouette score and cleaner t-SNE/UMAP projections.

B. Why Isolation Forest Diverges

The moderate agreement between reconstruction-based methods and IF ($\rho \approx 0.55$–0.57) deserves some reflection. Isolation Forest operates directly on the original 11-dimensional feature space using axis-aligned random splits, which makes it sensitive to features with high variance or heavy tails—in this case, `log_enrollment` and `student_teacher_ratio`. The autoencoders, by contrast, learn a joint representation of all features and penalize deviations from the multivariate pattern rather than from individual feature extremes. As a

Table VII: Illustrative Profiles of Top-10 AE Anomalous Schools (anonymized). STR = student-to-teacher ratio. Dom. = dominant ethnic group proportion.

Rank	Type	Enroll.	STR	FRPL%	Anomaly Driver
1	Virtual	4,210	183	62%	Extreme STR + large size
2	Charter	312	14	3%	99% single ethnicity + low FRPL
3	Public	3,801	22	88%	Very large + high poverty
4	Virtual	2,950	156	71%	Extreme STR
5	Charter	89	9	97%	Tiny size + 100% single ethnicity
6	Public	445	18	1%	Near-zero FRPL, 98% single ethnicity
7	Alternative	203	41	91%	High STR for brick-and-mortar
8	Public	3,540	19	45%	Outlier enrollment size
9	Charter	176	11	94%	100% single ethnicity + high FRPL
10	Virtual	1,870	140	58%	Extreme STR, imputed staff FTE

consequence, IF tends to surface schools that are unusual in one or two dimensions independently, while the AE/VAE flag schools whose *combination* of features is unusual, even if each individual feature is unremarkable.

A concrete example from the top-50 lists illustrates this: a medium-sized charter school with average enrollment and a typical demographic composition but a student-to-teacher ratio of 140 was flagged by IF (large path-length split on that one feature) but ranked only in the 75th anomaly percentile by the AE. Conversely, a school with perfectly median enrollment and teacher ratio but an unusual combination of 95% FRPL and 98% single-ethnicity enrollment was ranked in the top-10 by both AE and VAE but missed by IF entirely. This complementarity reinforces the practical recommendation to ensemble multiple detectors when the downstream cost of false negatives is high.

C. Threshold Sensitivity

Throughout this work we anchor anomaly labeling at the 95th-percentile threshold, yielding a 5% anomaly rate. This choice is intentionally conservative: in a dataset of 90,000 schools, 5% implies roughly 4,500 flagged institutions, which is already a large number to investigate manually. Relaxing the threshold to the 99th percentile reduces the flagged pool to approximately 900 schools, substantially increasing precision but risking under-coverage of borderline outliers. We explored both thresholds and observed that the top-500 anomaly rankings are effectively identical between AE and VAE regardless of the choice, suggesting that the high-confidence anomalies are robust to threshold selection. The 5% threshold was ultimately retained for consistency with the

IF contamination parameter and to enable direct three-way comparisons.

.8. Cross-Challenge Synthesis

This work builds directly on the clustering archetypes discovered in Challenge 5. The six K-means clusters revealed that U.S. public schools stratify primarily along racial/socioeconomic axes: high-poverty urban schools with majority-minority enrollment (Clusters 0, 1), affluent suburban White-majority schools (Cluster 2), charter-intensive urban schools (Cluster 3), rural schools with high Native American enrollment (Cluster 4), and suburban multi-ethnic schools (Cluster 5).

The AE/VAE framework adds two new insights:

(1) Intra-cluster anomalies. Within each archetype, there exist schools whose feature profiles are reconstructed poorly, suggesting they are outliers *within* their cluster—a dimension that K-means silhouette analysis cannot detect. These within-cluster anomalies may represent schools undergoing demographic transition, data entry errors, or genuinely exceptional institutions.

(2) Soft manifold structure. The VAE latent space shows that cluster boundaries are not hard partitions but gradient transitions, particularly between the two high-poverty clusters and between the charter and urban-public archetypes. This smooth geometry is more faithful to the underlying continuum of school characteristics than the discrete K-means assignment.

A. Policy Implications of Within-Cluster Anomalies

The intra-cluster anomaly perspective is arguably the most actionable finding for educational stakeholders. When a school is flagged as anomalous at the national level, it may simply reflect the fact that it belongs to an under-represented archetype—for example, a rural tribal school, which is statistically rare in absolute terms but behaves consistently with other tribal schools. Such schools should not necessarily be treated as problematic; they are anomalous only with respect to the majority distribution. However, when a school is flagged as an anomaly *within* its own archetype (i.e., relative to the subset of schools that share its demographic profile), the signal is harder to explain away as mere minority-group representation. These within-cluster anomalies—which we estimate constitute roughly 40% of the top-1,000 flagged schools—warrant more targeted follow-up. They may indicate schools that have recently changed operational structure (e.g., converted to charter status mid-year), experienced a data reporting error at the district level, or, more positively, adopted an unusual but effective staffing model.

B. Connecting All Three Challenges

Together, the three challenges in this project constitute a layered analytical pipeline that progressively deepens the characterization of U.S. public schools. Challenge 2 established a supervised baseline, demonstrating that school-level features carry enough predictive signal to distinguish school types with moderate accuracy. Challenge 5 moved to an unsupervised paradigm, revealing that demographic and socioeconomic features naturally organize schools into six interpretable archetypes without any label supervision. Challenge 6 closes the loop by asking not just *what are the typical patterns* but *which schools deviate from them*, and, through the latent-space analysis, *how can we represent schools in a lower-dimensional space that respects their structural similarity*.

This progression mirrors a common arc in applied machine learning projects: supervised models provide a ceiling on predictive performance, clustering models reveal latent structure, and anomaly detection models surface exceptions that challenge the taxonomy. Each layer informs the next—the cluster labels from C5 serve as external validation for the C6 latent space, and the anomaly scores from C6 could, in future work, be fed back as additional features or as soft labels for a weakly supervised model.

.9. Limitations and Ethical Considerations

A. Data Quality and Reporting Bias

The CCD dataset is an administrative record compiled from state-level submissions, which means its accuracy depends entirely on how consistently individual states and districts complete the reporting templates. Several of the most extreme anomalies we identified are better characterized as data artifacts than as genuinely unusual schools. For instance, two schools in the top-10 AE anomaly list have a recorded `student_teacher_ratio` above 180, which is almost certainly a data entry error (e.g., a part-time FTE reported as 0.01 rather than a full FTE). Our preprocessing pipeline clips this feature at 200, which prevents these entries from completely dominating the score distribution, but they still appear in the extreme tail. Downstream users of these anomaly scores should apply domain knowledge to distinguish methodological outliers from genuine ones.

Furthermore, several features in the dataset have non-trivial missing rates. Staff FTE counts, in particular, are missing for approximately 8% of schools in the raw merged file. Median imputation—while standard practice—introduces a systematic bias toward the median student-to-teacher ratio for these schools, which may artificially reduce their anomaly scores and cause genuinely anomalous schools with missing staff data to be under-flagged. A more robust imputation strategy, such as k-nearest-neighbor imputation conditioned on school level and locale, is a promising direction for future preprocessing work.

B. Ethical Considerations in Anomaly Labeling

Labeling a school as anomalous in a machine learning context is a statistical statement: the school's feature vector is distant from the majority of the training distribution. It carries no inherent normative meaning. However, in practice, anomaly scores derived from educational datasets can be misinterpreted or misused. A school that serves a small, geographically isolated community may receive a high anomaly score simply because its demographic profile is uncommon at the national level, not because there is anything wrong with how it operates. If such scores were used to inform resource allocation, audit prioritization, or accountability decisions, the statistical "anomaly" label could translate into concrete disadvantage for already-marginalized communities.

We want to be explicit that the anomaly scores produced in this work are intended as exploratory analytical tools, not as ranking systems for school quality or performance. The CCD data contains no outcome variables (test scores, graduation rates, safety records), and our models make no claims about whether anomalous schools are performing well or poorly. Any policy application of these scores would require careful integration with outcome data, expert review, and community input before any institutional conclusions are drawn.

A related concern is the choice of 5% contamination, which labels approximately 4,500 schools as anomalous. Because certain school types (virtual, charter, tribal, and alternative schools) are structurally different from the majority of brick-and-mortar public schools, they are disproportionately represented in the flagged pool. This is not a failure of the models—it is a direct reflection of how different these school types are statistically—but it does mean that aggregate anomaly rates should not be compared across school types without controlling for this structural confound.

.10. Conclusions

We presented a comprehensive unsupervised analysis of the NCES CCD 2022–23 dataset using AutoEncoders, Variational AutoEncoders, and Isolation Forest for anomaly detection and representation learning. Key findings:

1) AE and VAE achieve near-identical anomaly rankings ($\rho = 0.93$), but the VAE latent space produces superior cluster separation (silhouette +0.035) due to its KL-regularized posterior.
2) Deep reconstruction-based detectors complement IF: moderate inter-method agreement ($\rho \approx 0.56$) implies that jointly flagging schools across methods increases robustness.
3) A 16-dimensional bottleneck optimally balances reconstruction fidelity and latent-space structure for this 11-feature dataset.
4) Anomalous schools cluster into interpretable categories: extreme demographic homogeneity, outsized enrollment, abnormal staffing ratios, and data quality artifacts.
5) Within-cluster anomaly analysis provides more actionable signals for policy stakeholders than global anomaly ranking alone.

Future work includes incorporating temporal data across multiple school years to detect structural drift, extending the VAE to a conditional architecture conditioned on state or district-level features, and exploring contrastive or self-supervised pre-training objectives to improve latent structure. More robust imputation strategies and the integration of school outcome variables would also substantially increase the interpretability and downstream utility of the anomaly scores.

On the methodological side, an interesting open question is whether a single unified model—such as a Deep SVDD [12] variant adapted for tabular data, or a normalizing flow model—could simultaneously achieve the anomaly detection performance of the IF and the representation quality of the VAE. The two objectives are not naturally aligned: anomaly detection benefits from sharp, discriminative boundaries, while representation learning benefits from smooth, continuous manifolds. Finding architectures that balance both goals on large-scale educational tabular data remains, in our view, the most technically interesting open problem at the intersection of deep generative models and educational data mining.

More broadly, this work illustrates both the promise and the responsibility that comes with applying unsupervised machine learning to public-sector data. The models are agnostic to the institutional and historical context that shapes the data they consume. A school that looks anomalous to a neural network may be functioning exactly as intended for its community—and it is the responsibility of the analyst, not the algorithm, to make that distinction. We hope the explicit treatment of limitations and ethical considerations in Section .9 serves as a useful template for future work in this space.

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